

Proximal Point Algorithms

Strong Convergence in Banach Spaces

Based on Pathak: “An Introduction to Nonlinear Analysis and Fixed Point Theory” (2018)

Nonlinear Analysis and Fixed Point Theory

Chapter 8.4 & Extensions

Outline I

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What Are Proximal Point Algorithms?

Definition

A **proximal point algorithm** is an iterative method for finding zeros of maximal monotone operators in Banach spaces via the **resolvent** operator:

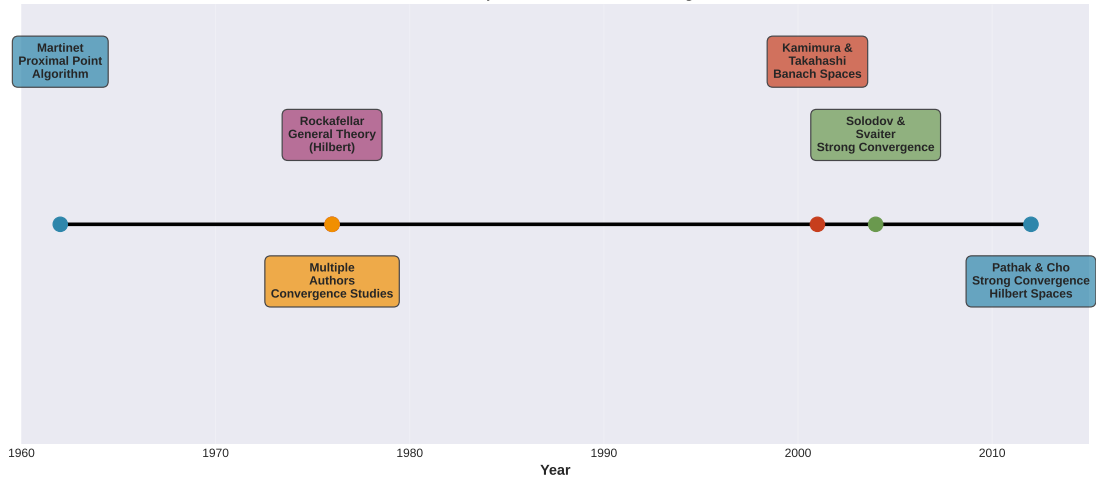
$$x_{n+1} = (I + r_n T)^{-1} x_n, \quad n = 0, 1, 2, \dots$$

where $T : X \rightarrow 2^X$ is a maximal monotone operator and $\{r_n\}$ is a sequence of positive parameters.

- Originally introduced by **Martinet (1962)** for finding zeros
- Developed by **Rockafellar (1976)** in general Hilbert space framework
- Extended to **Banach spaces** by multiple authors
- Modified for **strong convergence** by Solodov-Svaiter, Kamimura-Takahashi
- Further improved by **Pathak and Cho (2012)**

Historical Development

Historical Development of Proximal Point Algorithms



Key Motivations

- ① **Weak vs Strong Convergence:** Classical proximal point converges weakly to zeros of maximal monotone operators, but not necessarily strongly
- ② **Banach Space Challenges:** Analysis in general Banach spaces requires sophisticated tools (duality, resolvents)
- ③ **Applications:** Monotone operators arise in:
 - Variational inequalities
 - Optimization problems
 - Nonlinear PDE and integral equations
- ④ **Strong Convergence:** Many applications need convergence in norm, not just weak convergence

Monotone and Maximal Monotone Operators

Definition

Let X be a Banach space with dual X^* . An operator $T : X \rightarrow 2^{X^*}$ is:

- **Monotone** if for all $x, y \in \text{dom}(T)$, $u \in Tx$, $v \in Ty$:

$$\langle u - v, x - y \rangle \geq 0$$

- **Maximal Monotone** if it is monotone and for all $x \in X$:

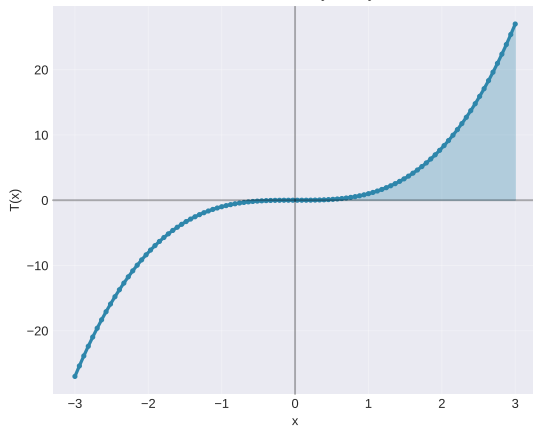
If $\langle v, x - y \rangle \geq 0$ for all $y \in \text{dom}(T)$, $v \in Ty$, then $v \in Tx$

Examples:

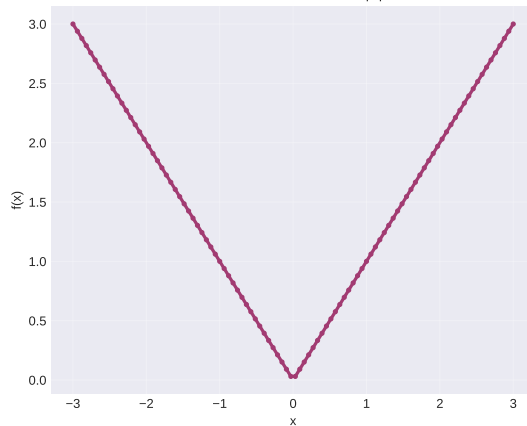
- Subdifferential of convex functions: $\partial f(x)$
- Duality mapping $J_p : X \rightarrow X^*$ (identity in Hilbert spaces)
- Negatives of monotone operators

Examples of Monotone Operators

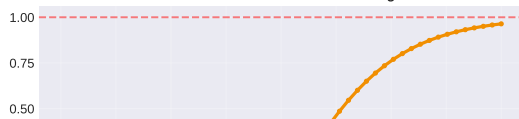
Monotone: $\langle T(x) - T(y), x - y \rangle \geq 0$



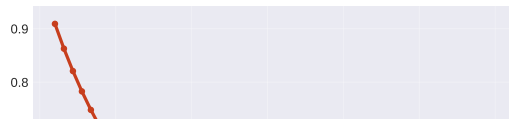
Subdifferential of $f(x) = |x|$



Maximal Monotone: Bounded Range



Resolvent $(I + rT)^{-1}$ Error vs r



The Resolvent and Its Properties

Definition

For a monotone operator T and $r > 0$, the **resolvent** is:

$$R_T^r = (I + rT)^{-1}$$

Theorem (Rockafellar 1976)

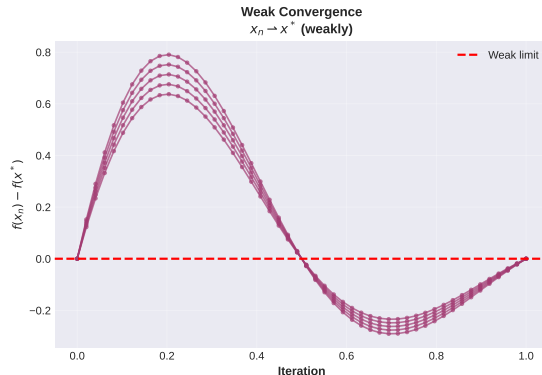
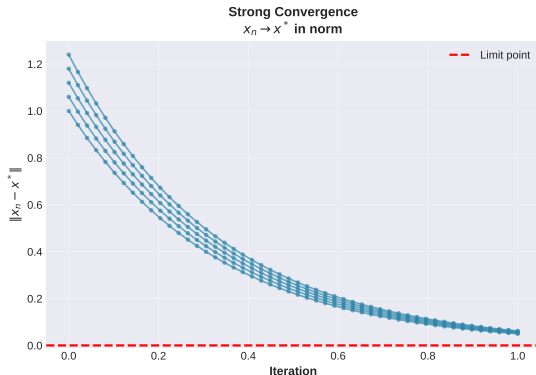
Let X be a reflexive, strictly convex and smooth Banach space, and let $T : X \rightarrow 2^{X^}$ be a monotone operator. Then T is maximal if and only if $R(J_p + rT) = X^*$ for all $r > 0$.*

Key Properties:

- Resolvent is **single-valued** and **nonexpansive**
- Fixed points: $\text{Fix}((I + rT)^{-1}) = T^{-1}0$
- Exists and is unique under appropriate conditions
- Error: $\|(I + rT)^{-1}x - x^*\| \rightarrow 0$ as iterations increase

- ④ Choose initial point $x_0 \in X$ and sequence $\{r_n\}_{n=0}^{\infty}$ with $r_n > 0$
- ② Set $n := 0$
- ③ **While** convergence criterion not satisfied:
 - ① Compute $x_{n+1} = (I + r_n T)^{-1} x_n$
 - ② Set $n := n + 1$

Weak vs Strong Convergence



- **Weak convergence** ($x_n \rightharpoonup x^*$): For all $f \in X^*$, $f(x_n) \rightarrow f(x^*)$
- **Strong convergence** ($x_n \rightarrow x^*$): $\|x_n - x^*\| \rightarrow 0$ in norm
- Strong convergence implies weak, but converse is false in infinite dimensions

The Solodov-Svaiter Algorithm (2001)

Solodov-Svaiter Modification

To generate strongly convergent sequences, Solodov and Svaiter **modified** the proximal point algorithm by incorporating half-space projections:

$$\begin{aligned}x_0 &\in X, \\ 0 &= v_n + \frac{1}{r_n}(y_n - x_n), \quad v_n \in T_{x_n}, \\ H_n &= \{z \in X : \langle z - y_n, v_n \rangle \leq 0\}, \\ W_n &= \{z \in X : \langle z - x_n, x_0 - x_n \rangle \leq 0\}, \\ x_{n+1} &= P_{H_n \cap W_n} x_0, \quad n = 0, 1, 2, \dots\end{aligned}$$

Key Idea: Use projections onto intersection of half-spaces to enforce strong convergence while maintaining the proximal point structure.

Kamimura-Takahashi Extension to Banach Spaces (2004)

Extension to Smooth Banach Spaces

In a smooth Banach space X , the modified proximal algorithm becomes:

$$\begin{aligned}x_0 &\in X, \\0 &= v_n + \frac{1}{r_n}(J_2(y_n) - J_2(x_n)), \quad v_n \in Tx_n, \\H_n &= \{z \in X : \langle z - y_n, v_n \rangle \leq 0\}, \\W_n &= \{z \in X : \langle z - x_n, J_2(x_0) - J_2(x_n) \rangle \leq 0\}, \\X_{n+1} &= P_{H_n \cap W_n} x_0, \quad n = 0, 1, 2, \dots\end{aligned}$$

where J_2 is the duality mapping with gauge function $t \mapsto t$.

Result: Strong convergence in uniformly smooth Banach spaces.

The Pathak-Cho Algorithm (2012) I

General Proximal-Type Algorithm in Smooth Banach Spaces

Let X be a smooth Banach space with duality mapping J_p . Consider:

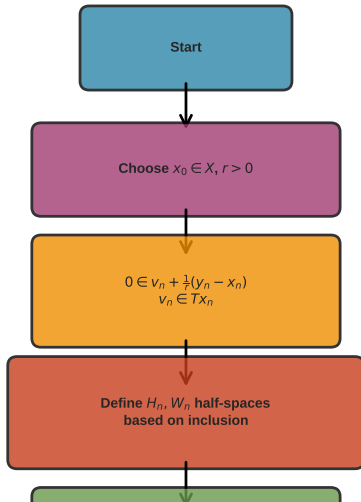
$$\begin{aligned}x_0 &\in X, \\0 &= v_n + \frac{1}{r_n}(J_p(y_n) - J_p(x_n)), \quad v_n \in Tx_n, \\H_n &= \{z \in X : \langle z - y_n, v_n \rangle \leq 0\}, \\W_n &= \{z \in X : \langle z - x_n, J_p(x_0) - J_p(x_n) \rangle \leq 0\}, \\x_{n+1} &= R_{H_n \cap W_n} x_0, \quad n = 0, 1, 2, \dots\end{aligned}$$

Notation:

- $R_{H_n \cap W_n}$ is the metric projection onto $H_n \cap W_n$
- J_p is the duality mapping with parameter p (generalization)
- The algorithm extends results of Kamimura-Takahashi to more general spaces

Algorithm Flowchart

Proximal Point Algorithm Flowchart



Fundamental Convergence Theorem

Theorem (Pathak and Cho 2012)

Let X be a reflexive, strictly convex and uniformly smooth Banach space. If $T^{-1}0 \neq \emptyset$, ϕ satisfies condition **(4.36)**, and $\{r_n\} \subset (0, \infty)$ satisfies $\liminf_{n \rightarrow \infty} r_n > 0$, then the sequence $\{x_n\}$ generated by the proximal-type algorithm **converges strongly** to $R_{T^{-1}0}x_0$.

Key Requirements:

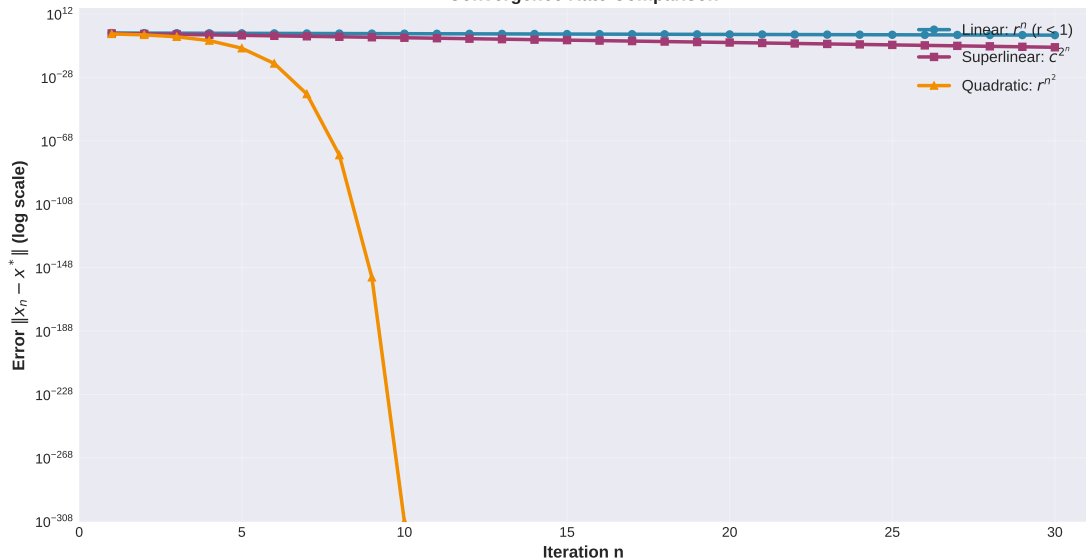
- 1 Reflexivity: Weak sequential compactness
- 2 Strict convexity: Unique projections
- 3 Uniform smoothness: Duality map properties
- 4 $T^{-1}0 \neq \emptyset$: Solution existence
- 5 Appropriate parameter selection

Convergence Analysis: Key Steps

- 1 **Well-Definedness:** Show that $H_n \cap W_n \neq \emptyset$ and projections exist
- 2 **Boundedness:** Prove $\{x_n\}$ and related sequences are bounded
- 3 **Monotonicity:** Establish a Lyapunov function $\Psi(x_n, x_0)$ that decreases monotonically
- 4 **Subsequential Convergence:** From compactness, extract weakly convergent subsequences
- 5 **Limit Identification:** Show limit is in $T^{-1}0$ using maximal monotonicity
- 6 **Full Convergence:** Use monotonicity of Ψ to extend from subsequences to full sequence

Convergence Rate Analysis

Convergence Rate Comparison



Convergence of the Sequence

Theorem

Let $\Psi(x_n, x_0)$ be defined as a Lyapunov function measuring distance between iterates. Then:

$$\Psi(x_n, x_{n+1}) + \Psi(x_{n+1}, x_0) \leq \Psi(x_n, x_0)$$

Since $\{\Psi(x_n, x_0)\}$ is monotone decreasing and bounded below, $\lim_{n \rightarrow \infty} \Psi(x_n, x_0)$ exists. From the inequality:

$$\Psi(x_n, x_{n+1}) \rightarrow 0, \quad \text{hence} \quad x_{n+1} - x_n \rightarrow 0$$

This combined with boundedness and weak sequential compactness gives the existence of convergent subsequences.

Sufficient Conditions for Strong Convergence I

Theorem

In a reflexive, strictly convex and uniformly smooth Banach space:

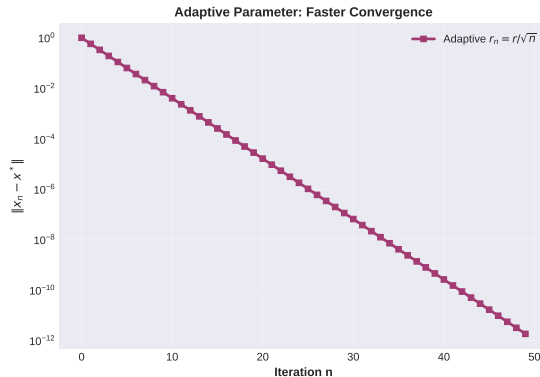
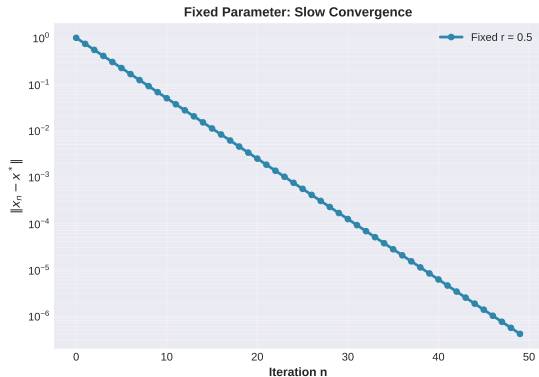
- ① *If T is a maximal monotone operator with $T^{-1}0 \neq \emptyset$*
- ② *And $\{r_n\}$ satisfies appropriate conditions (e.g., bounded away from 0)*
- ③ *Then the sequence generated by the proximal algorithm exhibits:*
 - *Boundedness of iterates*
 - *Monotone decrease of Lyapunov function*
 - *Strong convergence to $P_{T^{-1}0}x_0$*

Sufficient Conditions for Strong Convergence II

Functional Framework:

- **Reflexive spaces:** x_n bounded $\Rightarrow$ subsequential weak convergence
- **Strictly convex:** Unique projections onto convex sets
- **Uniformly smooth:** Duality map is uniformly continuous, enabling strong estimates
- **Maximal monotone:** Guarantees resolvent properties and surjectivity conditions

Parameter Selection Strategies

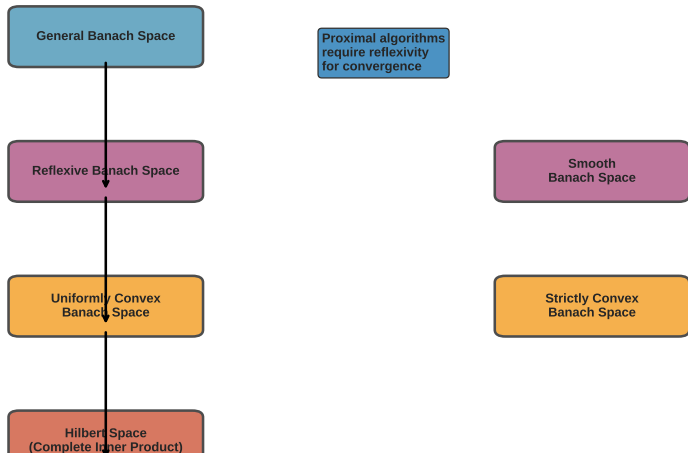


Effective Choices for $\{r_n\}$:

- Fixed: $r_n = r > 0$ (simplest, but slower)
- Decreasing: $r_n = r_0/\sqrt{n+1}$ (faster convergence)
- Adaptive: $r_n = r_0 \cdot f(n)$ with $f(n) \rightarrow 1$ as $n \rightarrow \infty$

Banach Space Hierarchy and Requirements

Banach Space Hierarchy and Proximal Algorithm Requirements



Why Specific Space Properties Matter

Reflexivity

Essential for: Weak sequential compactness of bounded sequences.

Without it: Cannot guarantee convergent subsequences from boundedness alone.

Strict Convexity

Essential for: Uniqueness of best approximations (projections).

Without it: Projection P_C may not be single-valued.

Uniform Smoothness

Essential for: Controlled behavior of duality mapping J_p .

Without it: Cannot establish sufficient contraction properties for convergence analysis.

Examples of Suitable Spaces

- ① **Hilbert Spaces:** Automatically reflexive, strictly convex, and uniformly smooth (with $p = 2$)
- ② ℓ^p spaces ($1 < p < \infty$): Reflexive and uniformly convex. Smooth for $p \geq 2$
- ③ $L^p(\Omega)$ spaces ($1 < p < \infty$): Banach lattices that are reflexive and uniformly convex
- ④ **Sobolev spaces** $W^{1,p}(\Omega)$: Suitable for PDE applications
- ⑤ **Orlicz spaces:** Generalizations allowing more flexibility

Not suitable: c_0 , ℓ^1 , $L^1(\Omega)$, $C(\Omega)$ (lack reflexivity or uniform convexity)

Applications: Variational Inequalities

Problem

Find $x \in K$ such that

$$\langle Tx, y - x \rangle \geq f(y) - f(x) \quad \forall y \in K$$

where K is a closed convex subset, T is a monotone operator, and f is convex.

Connection to Proximal Algorithms:

- The variational inequality is equivalent to:

$$0 \in (T + \partial f + N_K)(x)$$

where N_K is the normal cone

- The composite operator $T + \partial f + N_K$ is maximal monotone
- Apply proximal algorithm to compute solutions iteratively

Application: Convex Optimization

Problem: Minimize convex function

```
# Minimize  $f(x)$  where  $f$  is convex  
# This is equivalent to finding  $x$  in  $\text{dom}(f)$  such that  
#  $0$  in  $df(x)$ , where  $df$  is the subdifferential of  $f$   
# The subdifferential is a maximal monotone operator!  
  
# Proximal algorithm:  
#  $x_{n+1} = (I + r_n * df)^{-1} x_n$   
  
# Example:  $f(x) = ||x||^2 / 2$   
def proximal_step(x_n, r_n):  
    #  $(I + r_n * I)^{-1} x_n = x_n / (1 + r_n)$   
    return x_n / (1.0 + r_n)
```

Application: Monotone Inclusion Problems

Problem

Find $x \in X$ such that $0 \in Tx$ where T is a maximal monotone operator.

Proximal algorithm solution:

- The classical proximal point algorithm directly applies
- For composite problems:

$$0 \in T_1x + T_2x + \cdots + T_mx$$

use splitting methods or proximal variants

- Strong convergence allows numerical verification of solutions
- Applications to PDEs:
 - Semilinear elliptic equations
 - Parabolic equations with subdifferential
 - Obstacle problems

Example 1: Simple Subdifferential Problem

Problem

Minimize $f(x) = |x|$ (absolute value function)

```
import numpy as np

def proximal_abs_minimize():
    # Subdifferential of |x|:
    # d|x| = {sign(x)} for x != 0
    # d|0| = [-1, 1]

    x = 10.0 # initial point
    r_n = [0.5 * (1 - 0.01*i) for i in range(50)]

    for n, r in enumerate(r_n):
        # Proximal step for | . |
        if abs(x) <= r:
            x = 0 # Shrinkage
        else:
            x = np.sign(x) * (abs(x) - r)

        print(f"Iter {n}: x = {x:.6f}")

    # Solution: x* = 0
    return x
```

Example 2: Quadratic Regularization

Problem

Minimize $f(x) = \|Ax - b\|^2/2 + \lambda\|x\|$ (Lasso problem)

```
import numpy as np
from scipy import linalg

def proximal_lasso(A, b, lam, x0, max_iter=100):
    m, n = A.shape
    x = x0.copy()

    for k in range(max_iter):
        # Gradient of quadratic part
        grad = A.T @ (A @ x - b)

        # Proximal step for L1 norm
        # prox_{lambda |.|} (x - grad) = soft_threshold
        x = soft_threshold(x - grad, lam)

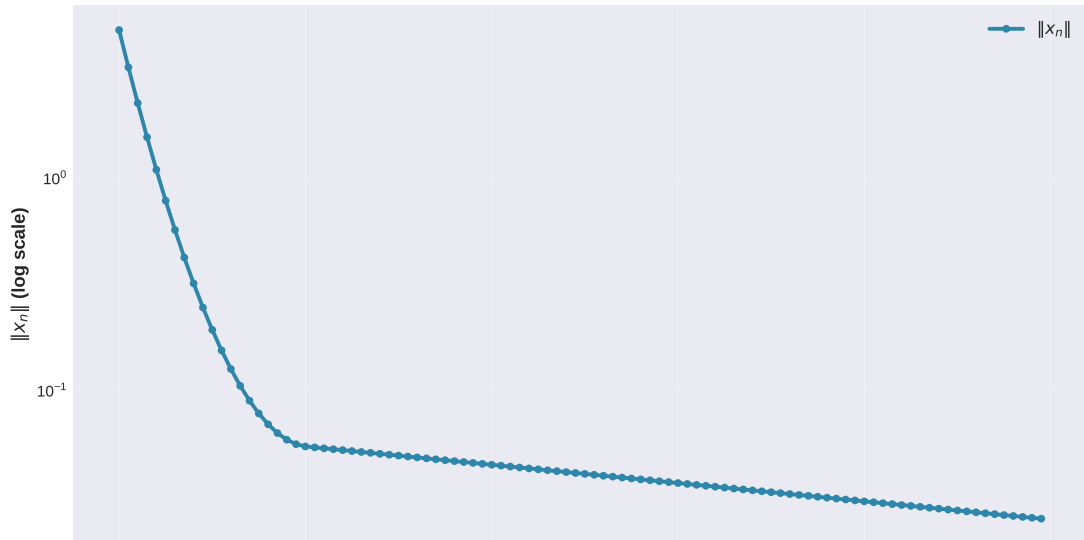
    return x

def soft_threshold(x, threshold):
    """Soft thresholding operator"""
    return np.sign(x) * np.maximum(np.abs(x) - threshold, 0)
```

Proximal Convergence Example

Convergence of Proximal Point Algorithm

$$x_{n+1} = (I + r_n T)^{-1} x_n$$



Occasions Pseudomonotone Operators

Definition

An operator $T : X \rightarrow 2^{X^*}$ is **occasionally pseudomonotone** if: For every $x \in X$ and sequences $\{x_n\}$, if $x_n \rightarrow x$ and $u_n \in Tx_n$ with $u_n \rightarrow u$, and $\limsup_n \langle u_n, x_n - x \rangle \leq 0$, then $u \in Tx$.

Significance:

- Weaker than monotonicity (allows more general operators)
- Still allows convergence analysis of proximal algorithms
- Includes multivalued mappings and subgradients
- Crucial for applications to nonlinear PDEs

Proximal Algorithms vs Other Methods

Method	Space	Convergence	Strong	Rate
Classical Proximal	H.S.	Weak	No	Linear
Solodov-Svaiter	H.S.	Strong	Yes	Linear
Kamimura-Takahashi	Ban.	Strong	Yes	Linear
Pathak-Cho	Ban.	Strong	Yes	Linear+
Gradient Descent	Any	Local	Yes	Linear
Newton's Method	Any	Local	Yes	Quadratic

Note: H.S. = Hilbert Space, Ban. = Banach Space

Summary of Main Results I

- ① **Classical Algorithm (1962-1976):** Weak convergence in Hilbert spaces
- ② **Solodov-Svaiter (2001):** First strong convergence via projections
- ③ **Kamimura-Takahashi (2004):** Extension to smooth Banach spaces
- ④ **Pathak-Cho (2012):** Generalization to uniform smoothness, applications
- ⑤ **Key Innovation:** Use of half-space projections $P_{H_n \cap W_n}$ instead of simple resolvent
- ⑥ **Unified Framework:** Works for:
 - Maximal monotone operators
 - Occasionally pseudomonotone operators
 - Operators with measurability properties

Key Takeaways

- **Proximal point algorithms** are powerful tools for solving monotone inclusion problems in Banach spaces
- **Strong convergence** is achievable through careful modifications (projections onto intersecting half-spaces)
- **Space structure** (reflexivity, convexity, smoothness) is essential for theoretical guarantees
- **Parameter selection** affects convergence rate; adaptive strategies can improve performance
- **Applications** range from optimization to solving nonlinear PDEs and variational inequalities

Future Research Directions

- ① **Splitting Methods:** Proximal algorithms for composite operators $T_1 + T_2 + \dots$
- ② **Stochastic Variants:** Proximal algorithms with noise/stochasticity for large-scale problems
- ③ **Accelerated Methods:** Combining proximal ideas with acceleration (Nesterov-type)
- ④ **Inverse Problems:** Applications to ill-posed problems and regularization
- ⑤ **Metric Space Generalizations:** Beyond Banach spaces to CAT spaces, manifolds
- ⑥ **Numerical Implementations:** Efficient implementations for sparse, large-scale problems

References and Further Reading

-  H. K. Pathak, *An Introduction to Nonlinear Analysis and Fixed Point Theory*, 2nd Edition, Springer, 2018.
-  R. T. Rockafellar, Monotone operators and the proximal point algorithm, *SIAM J. Control Optimization*, 14(5), 1976.
-  M. V. Solodov and B. F. Svaiter, A hybrid projection-proximal point algorithm, *J. Convex Analysis*, 6, 2001.
-  S. Kamimura and W. Takahashi, Strong convergence of a proximal-type algorithm in a Banach space, *SIAM J. Optimization*, 13(4), 2004.

Thank You!

Questions and Discussion